

A Hybrid LLM and Rule-Based Framework for Transparent Clinical Trial Eligibility Matching

Vasco Rodrigues
DEI
University of Coimbra
Coimbra, Portugal
vascorodrigues1243@gmail.com

Pedro Nuno Furtado
CISUC, DEI
University of Coimbra
Coimbra, Portugal
pnf@dei.uc.pt

Margarida Marques
DTIP
Local Health Unit of Coimbra (ULS Coimbra)
Coimbra, Portugal
margaridamarques@ulscoimbra.min-saude.pt

Adelino Pais
DTIP
Local Health Unit of Coimbra (ULS Coimbra)
Coimbra, Portugal
adelino.pais@ulscoimbra.min-saude.pt

Judy Paulo
Local Health Unit of Coimbra (ULS Coimbra)
Faculty of Medicine, University of Coimbra
Coimbra, Portugal
13097@ulscoimbra.min-saude.pt

Abstract—Identifying eligible patients for oncology clinical trials remains a time-consuming task due to the large volume of unstructured clinical documentation and the complexity of eligibility criteria. Although Large Language Models (LLMs) have shown considerable promise for supporting this process, most existing approaches rely primarily on probabilistic reasoning, limiting transparency and reproducibility.

This paper presents a clinician-validated hybrid framework that combines LLM-based information extraction with deterministic rule-based eligibility assessment. Unstructured oncology records are transformed into structured patient profiles, while clinical trial eligibility criteria are extracted and converted into machine-interpretable representations. Patient eligibility is then assessed using deterministic evaluation whenever possible, with LLM-assisted reasoning reserved for criteria that cannot be formally represented.

The proposed framework was evaluated using three open-source 8B LLMs across five pipeline tasks. The best-performing model achieved 92.3% accuracy in eligibility criteria extraction, 78.3% in patient profile extraction, and up to 84.0% accuracy in patient-trial matching. These results demonstrate that combining deterministic reasoning with lightweight LLMs provides a transparent and clinically reliable approach for supporting oncology trial recruitment.

Index Terms—Large Language Models; Information Extraction; Clinical Trial Eligibility; Unstructured Clinical Data; Clinician-in-the-Loop; Semi-Deterministic Eligibility Evaluation; Structured Patient Data

INTRODUCTION

Oncology clinical trials remain essential for advancing cancer treatment, yet a substantial proportion of eligible patients are never enrolled due to the complexity of manually assessing eligibility from unstructured electronic health records. Clinicians must review lengthy clinical notes and compare them against extensive trial protocols, making patient recruitment a time-consuming and error-prone process.

Recent advances in Large Language Models (LLMs) have shown considerable promise for both clinical information extraction and patient-trial matching. However, most existing approaches rely predominantly on probabilistic reasoning or assume the availability of structured electronic health records, limiting transparency, reproducibility, and clinical auditability. Furthermore, information extraction and eligibility assessment are often treated as independent tasks rather than components of a unified clinical workflow.

This paper presents a clinician-validated hybrid framework that combines LLM-based information extraction with deterministic rule-based eligibility evaluation. The proposed pipeline transforms unstructured oncology notes into structured patient profiles, extracts and formalizes eligibility criteria from clinical trial protocols, and performs transparent patient-trial matching by combining logical rule execution with LLM-assisted reasoning only when deterministic evaluation is not possible.

The framework was evaluated using three open-source 8B-parameter LLMs across five pipeline tasks. The best-performing models achieved 92.3% accuracy for eligibility extraction, 78.3% for patient profile extraction, and up to 84.0% in patient-trial matching while producing clinician-rated justification scores of 4.93/5. These results demonstrate that combining deterministic reasoning with lightweight LLMs can support transparent and clinically reliable decision support for oncology trial recruitment.

RELATED WORK

Recent advances in Large Language Models (LLMs) have significantly improved clinical natural language processing, particularly for extracting structured information from unstructured electronic health records [11], [12]. Several studies have demonstrated that instruction-tuned LLMs can identify

diagnoses, treatments, biomarkers, and other clinically relevant entities with high accuracy [1], [3], [9]. Nevertheless, these approaches typically focus on information extraction as an isolated task and rarely integrate the extracted information into complete clinical decision-support workflows [9], [11].

Clinical trial matching has also benefited from recent advances in LLMs. Existing approaches commonly rely on semantic similarity, retrieval-augmented generation, or end-to-end probabilistic reasoning to assess patient eligibility [4], [8], [12]. While these methods achieve encouraging results, they often lack transparency and make it difficult for clinicians to verify how individual eligibility decisions are reached [4], [12]. Furthermore, many assume structured electronic health records, limiting their applicability in routine clinical practice [4], [6].

With these gaps in mind, the proposed framework combines LLM-based information extraction with deterministic rule evaluation and clinician validation throughout the workflow. Rather than relying exclusively on probabilistic reasoning, eligibility assessment is performed deterministically whenever possible, with LLM-assisted reasoning reserved for criteria that cannot be formally represented. This hybrid design builds upon previous hybrid approaches [3], [4] while explicitly separating structured information extraction from eligibility reasoning, improving transparency, traceability, and clinician oversight throughout the complete patient-trial matching workflow.

PROPOSED FRAMEWORK

The proposed framework aims to reduce the workload associated with patient screening by leveraging LLMs to extract clinically relevant information from free-text oncology records and support transparent clinical trial eligibility assessment. Rather than relying exclusively on probabilistic model decisions, the framework incorporates clinician validation throughout the pipeline, ensuring that all structured patient profiles and eligibility criteria are reviewed before being used for patient matching.

The workflow is organized into three interconnected phases:

- Phase 1: Hyper-Relevant Parameter Extraction;
- Phase 2: Extraction and Conversion of Clinical Criteria;
- Phase 3: Semi-Deterministic Trial-Patient Matching.

Hyper-Relevant Parameter Extraction

The first phase transforms free-text oncology clinical notes into normalized structured patient profiles. Clinical diaries selected by the clinician are first pre-processed to remove irrelevant content before being provided to the LLM together with a task-specific prompt.

The model extracts clinically relevant parameters according to a predefined patient schema while simultaneously normalizing diagnoses using a stakeholder-defined mapping. The generated patient profile is then presented to the clinician

for validation and possible corrections. Once approved, the structured profile is stored and becomes available for the remaining stages of the pipeline.

A formalization of this workflow can be seen on the Figure 1 below.

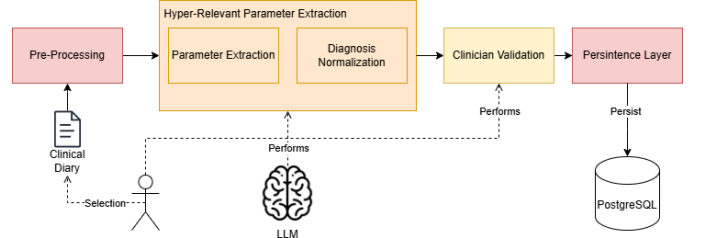


Fig. 1. Hyper-Relevant Parameter Extraction

Extraction and Conversion of Clinical Criteria

The second phase converts clinical trial eligibility requirements into machine-interpretable representations suitable for automated evaluation. The process begins by pre-processing the clinical trial document, removing sections unrelated to eligibility assessment. The LLM then identifies the trial structure, including the existence of different cohorts, providing additional context for the extraction stage.

Using this information, the model extracts inclusion and exclusion criteria while preserving their association with the corresponding cohort whenever applicable. Individual criteria are extracted whenever possible, although compound criteria are preserved when necessary to maintain their clinical meaning. The extracted criteria are subsequently validated by the clinician before being translated into structured logical expressions. These logical representations are again reviewed by the clinician to ensure that they accurately capture the original intent of the eligibility criteria before becoming available for patient matching.

A formalization of this workflow can be seen on the Figure 2 below.

Semi-Deterministic Trial-Patient Matching

The final phase performs patient-trial matching using the validated artifacts produced by the previous stages. Structured patient profiles generated during Phase 1 are combined with the validated logical eligibility criteria obtained during Phase 2 to identify potentially eligible patients for a selected clinical trial.

After selecting a trial, the system retrieves a subset of patient profiles based on the pathology group defined during trial registration, reducing unnecessary comparisons and LLM calls. Each patient is then evaluated against every eligibility criterion using one of two complementary assessment strategies:

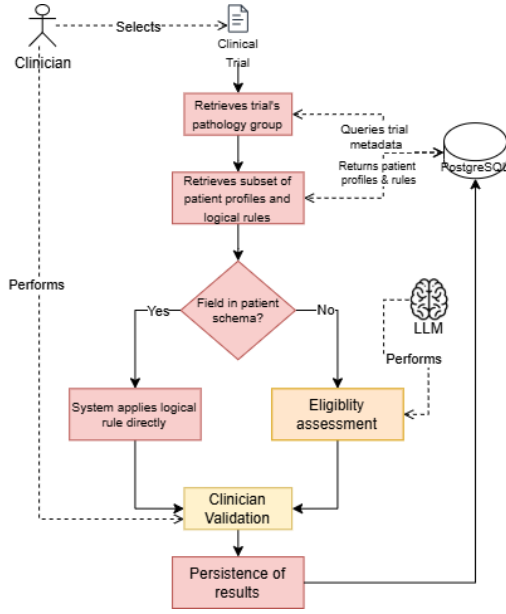


Fig. 2. Extraction and Conversion of Clinical Criteria Workflow

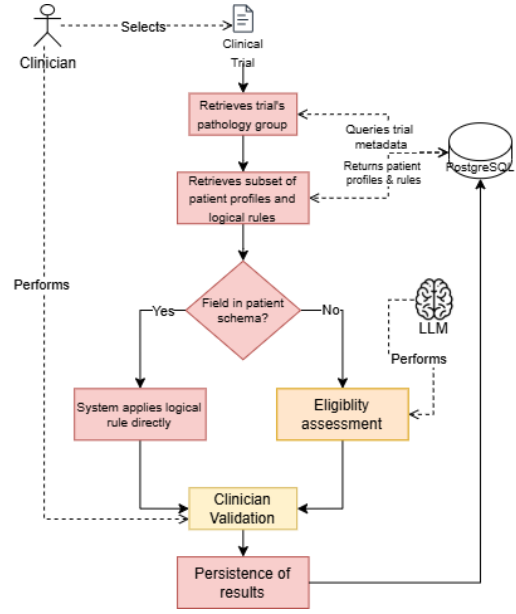


Fig. 3. Semi-Deterministic Trial-Patient Matching

- *Deterministic Assessment* - if the criterion references a field available in the patient profile schema, the corresponding logical rule is evaluated directly against the structured patient data;
- *LLM-Assisted Assessment* - if the criterion references information that is not represented within the structured schema, the LLM receives the patient profile, clinical diary, laboratory results, and the corresponding criterion to determine eligibility. In addition to the assessment result the model provides a justification to support expert review.

Although different evaluation mechanisms are employed, both strategies produce the same output: a criterion-level eligibility assessment. The complete set of assessments is then presented to the clinician for review, allowing any automatically generated decision to be confirmed or modified. Finally, all criterion evaluations, manual corrections, and the resulting eligibility decision are stored for traceability and future auditing.

A formalization of this workflow can be seen on the Figure 3 below.

EXPERIMENTAL EVALUATION

This section describes the experimental setup used to evaluate the proposed framework, including the datasets, evaluated models, and the obtained results.

Datasets

The proposed framework was evaluated using two datasets. The first consists of 30 synthetic free-text oncology clinical diaries generated with the open-source `gpt-oss-120b` model.

The generation process was grounded on four representative real clinical diaries, which served as references for writing style, structure, and clinical content, allowing the synthetic documents to reproduce the variability and noise commonly found in real-world clinical documentation.

The second dataset comprises 30 clinical trials collected from *ClinicalTrials.gov*. The selected studies cover a diverse range of tumour types, disease stages, molecular profiles, and therapeutic settings, providing representative eligibility criteria for evaluating the proposed pipeline.

Models

Three open-source Large Language Models from different model families were evaluated:

- Qwen3 8B Instruct;
- Llama 3.1 8B Instruct;
- DeepSeek-R1 8B.

All experiments were performed using deterministic inference settings to ensure reproducibility and minimize output variability. No fine-tuning was performed, and all models were evaluated under a zero-shot prompting strategy.

Results

The proposed framework was evaluated across the five tasks that compose the complete patient-trial matching pipeline. Overall, the results reveal a consistent performance pattern across all evaluated models. Structured extraction tasks achieved the highest accuracies, while translating free-text eligibility criteria into formal logical representation proved to be the most challenging stage. Nevertheless, LLM-assisted

reasoning enabled the framework to recover performance during the final patient-trial matching task.

Table I summarizes the best-performing model for each evaluated task.

TABLE I
PIPELINE PERFORMANCE ACROSS TASKS

Task	Best Model	Accuracy
Patient Profile Extraction	Qwen3	78.3%/83.3%
Trial Structure Identification	DeepSeek	96.7%/96.7%
Criteria Extraction	Qwen3	92.3%/95.9%
Criteria Extraction - cohort specific	Qwen3	96.3%/96.3%
Criteria Conversion	DeepSeek	64.1%/73.4%
Patient Matching	Qwen3	84.0%/84.0%

Hyper-Relevant Parameter Extraction: Patient profile extraction achieved consistently high performance across all evaluated models, with Qwen3 obtaining the highest strict accuracy of 78.3%. All models maintained 100% schema compliance, confirming their ability to reliably generate structurally valid JSON outputs. Qwen3-8B also achieved 83.3% partial accuracy and the lowest missing field rate (11.6%), followed by DeepSeek-r1:8b with 74.3% strict accuracy, 80.5% partial accuracy, and a 13.3% missing rate. Llama3.1-8b showed weaker performance, with 62.3% strict accuracy, 70.3% partial accuracy, and a substantially higher missing field rate of 21.7%. Explicit fields such as Gender and ECOG-PS reached 100% accuracy across all models, whereas diagnosis normalization, hierarchical treatment extraction, full TMN staging, and date-related fields remained the primary sources of error. These challenges were largely associated with temporally complex treatment histories, normalization requirements, and multi-level clinical structures rather than failures in identifying individual clinical entities.

The Figure 4 presents a visualization of the results.

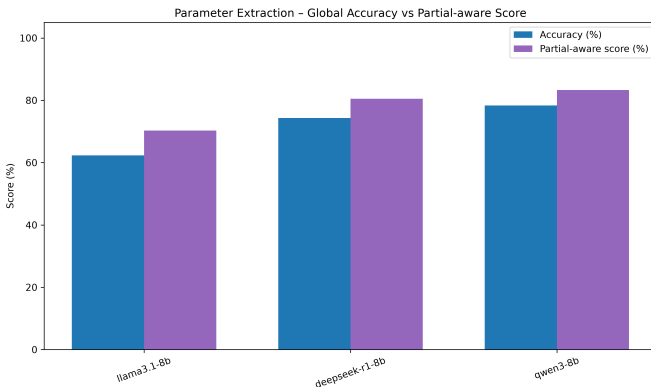


Fig. 4. Hyper-Relevant Parameter Extraction - strict vs partial accuracy

Trial Structure Identification: Trial Structure Identification was evaluated considering: (i) identification of whether formal cohort structures were present, and (ii) correctness of the generated cohort representation. Substantial differences were observed between models. DeepSeek-r1:8b achieved the best

performance, reaching 96.7% strict accuracy, while Llama3.1-8b presented a considerably lower result of 25% strict accuracy, mainly due to difficulties distinguishing formal trial cohorts from descriptive patient groups. Qwen3-8b obtained intermediate performance, with 60% strict accuracy and a 75% partial score. Models with weaker results showed recurrent issues both in misinterpreting descriptive clinical groups as cohorts and in complying with the enforced schema. Expert evaluation further indicated that correct cohort identification does not necessarily guarantee preservation of clinical meaning. Across the analyzed trials, the predominant errors were related to omission of eligibility-defining information rather than incorrect cohort structures.

The Figure 5 presents a visualization of the results.

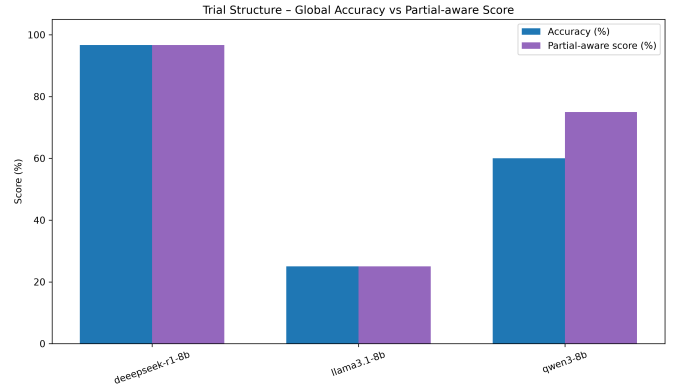


Fig. 5. Trial Structure Identification - strict vs partial accuracy

Eligibility Criteria Extraction: Eligibility criteria extraction evaluation was performed in two different tasks: (i) extraction of individual inclusion and exclusion criteria, and (ii) identification of whether each criterion belonged to a specific trial cohort. The first task was the strongest-performing component of the pipeline, with Qwen3-8b achieving 92.3% strict accuracy and 95.9% partial accuracy. DeepSeek-r1:8b also achieved competitive results, with 83.7% strict and 88.2% partial accuracy, while Llama3.1-8b obtained 71.9% strict and 75.8% partial accuracy, generating the highest number of incorrect criteria. The second task evaluated cohort attribution, where Qwen3-8b again achieved the best performance with 96.3% accuracy, whereas both DeepSeek-r1:8b and Llama3.1-8b reached 66.7% accuracy, with most errors arising from incomplete extraction rather than incorrect cohort classification. Expert review further confirmed these trends: Qwen3-8b consistently preserved the clinical meaning of extracted criteria, even when restructuring compound requirements; DeepSeek-r1:8b maintained high semantic fidelity and avoided hallucinations; and Llama3.1-8b showed more variable behavior, occasionally fragmenting compound criteria in ways that altered their logical structure. Overall, deviations across models were primarily related to representation differences, such as omission of eligibility components or incorrect fragmentation, rather than failures in clinical interpretation, reinforcing the importance

of clinician validation when transforming natural-language eligibility requirements into computational representations.

The Figures 6 and 7 presents a visualization of the results.

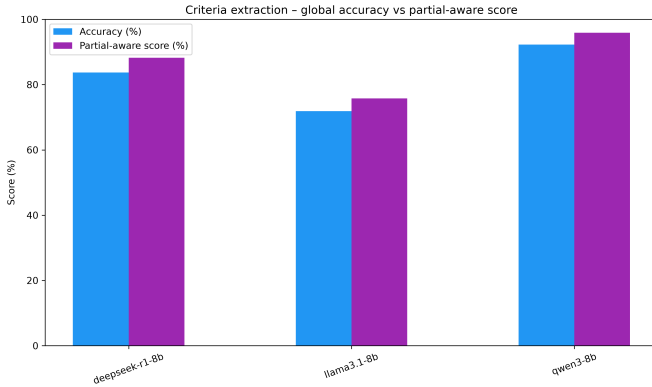


Fig. 6. Eligibility Criteria Extraction - strict vs partial accuracy

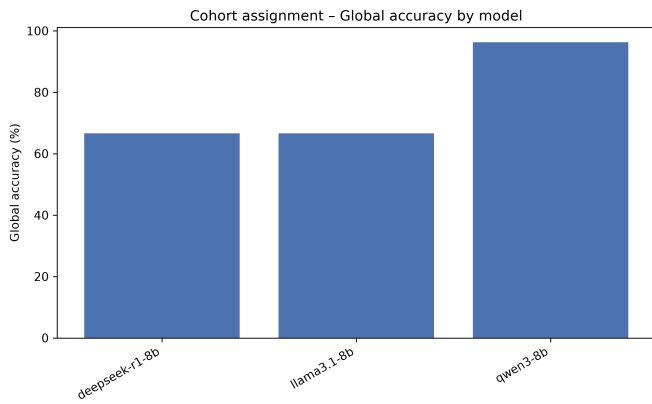


Fig. 7. Eligibility Criteria Extraction - Cohort assignment strict accuracy

Eligibility Criteria Conversion: Converting natural-language eligibility criteria into formal logical expressions remained the most challenging component of the pipeline. The best-performing 8B model was Qwen3-8b, reaching 65.1% strict accuracy and 74.4% partial accuracy, although it also produced the highest number of incorrect conversions. Llama3.1-8b achieved comparable results, with 62.5% strict and 73.5% partial accuracy, while DeepSeek-r1:8b presented the lowest strict accuracy (52.2%) but a substantially higher partial score (71.2%).

Across all evaluated models, a consistent failure pattern involved difficulty converting complex compound criteria. Eligibility requirements containing nested logical relationships—particularly combinations of multiple AND/OR conditions—frequently resulted in partially represented rules, with models extracting only the most prominent condition while omitting secondary constraints or simplifying relationships between conditions. Another recurrent issue involved incorrect mapping of criteria to existing schema fields, especially treatment-related constraints. Models often failed to recognize

that certain criteria could be directly represented using the patient schema, instead generating new non-schema fields that would unnecessarily force LLM-based evaluation downstream.

Given that this step represented the bottleneck of the pipeline, a larger model was evaluated to determine whether increased capacity improved conversion quality. Gpt-oss-120b achieved the best overall results, with 65.1% strict accuracy and 80.9% partial accuracy, and a substantially lower number of incorrect conversions (35) compared to the smaller models. Notably, most of its partial conversions did not arise from failures in capturing complex logical structures, but rather from incorrect normalization of criteria fields to the defined schema—again most prominently in treatment-related constraints.

The Figures 8 presents a visualization of the results.

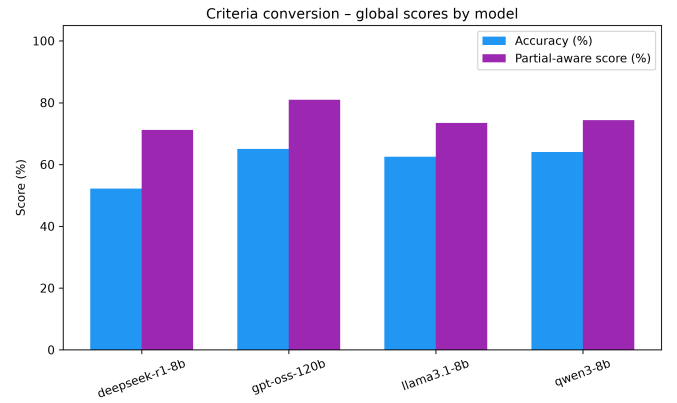


Fig. 8. Eligibility Criteria Conversion - strict vs partial accuracy

Patient-Trial Matching: Despite the lower performance observed during logical formalization, the final patient-trial matching stage presented substantially better results. Qwen3-8b achieved the highest strict accuracy of 84.0%, closely followed by DeepSeek-r1:8b with 83.0%, while Llama3.1-8b obtained 78.0%. No partially correct classifications were observed, as incorrect decisions typically resulted from misinterpretation of clinical evidence rather than partial satisfaction of the criterion.

Expert evaluation further confirmed these trends. Qwen3-8b achieved the highest justification score (4.93/5), consistently grounding decisions in explicit patient information and adopting conservative reasoning when data were missing. DeepSeek-r1:8b also demonstrated strong reliability, with an expert score of 4.8/5, correctly distinguishing between absent and unavailable information and avoiding unsupported assumptions. Llama3.1-8b obtained 4.2/5, performing well when information was explicit but occasionally over-interpreting uncertainty, such as rejecting criteria despite adequate laboratory values or inferring conclusions from incomplete documentation. Overall, the expert assessment showed that model reliability in this stage depended not only on accuracy but also on transparent, evidence-based justification, reinforcing

the importance of clinician-in-the-loop validation.

The Figures 9 presents a visualization of the results.

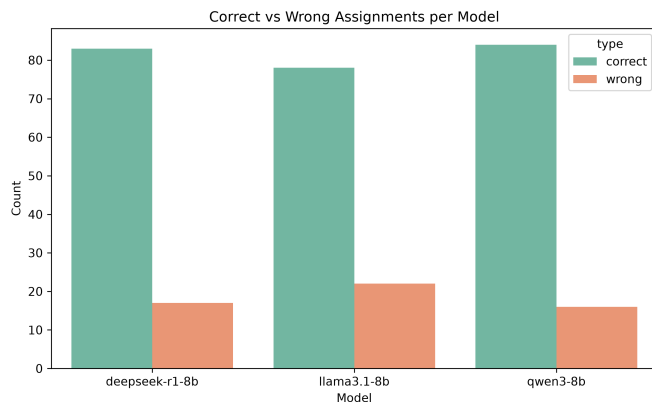


Fig. 9. Patient-Trial Matching - correct vs wrong

DISCUSSION

The experimental results demonstrate that lightweight open-source LLMs are capable of supporting multiple stages of a clinical trial matching workflow, although performance varies according to task complexity. Across all evaluated models, structured extraction tasks consistently achieved the highest accuracies, while transforming natural-language eligibility criteria into formal logical representations remained the principal bottleneck of the pipeline.

The results also highlight that no single model consistently outperformed the others across all tasks. Qwen3 achieved the best performance in patient profile and eligibility criteria extraction, whereas DeepSeek excelled in trial structure identification and demonstrated robust reasoning during patient-trial matching. These observations suggest that decomposing the workflow into specialized components enables each task to benefit from the strengths of different models, rather than relying on a single end-to-end solution.

Finally, the proposed hybrid strategy combines deterministic rule evaluation with LLM-assisted reasoning only when structured information is insufficient. Together with clinician validation at every stage, this design improves transparency and traceability while reducing dependence on purely probabilistic model decisions, characteristics that are essential for trustworthy clinical decision-support systems.

CONCLUSION

This paper presented a clinician-validated hybrid framework for semi-deterministic patient-trial matching from unstructured oncology clinical documentation. The proposed methodology combines LLM-based information extraction, logical formalization of eligibility criteria, deterministic rule evaluation, and LLM-assisted reasoning to support transparent clinical trial recruitment.

The experimental evaluation showed that lightweight open-source LLMs are capable of performing clinically relevant tasks across the proposed pipeline. Structured extraction achieved the strongest results, with Qwen3 reaching 78.3% strict accuracy for patient profile extraction and 92.3% for eligibility criteria extraction, while patient-trial matching achieved up to 84.0% accuracy. In contrast, logical formalization remained the most challenging stage, highlighting an important direction for future research.

Overall, the proposed framework demonstrates that combining structured representations, deterministic reasoning, and clinician supervision provides a practical and explainable approach for supporting oncology clinical trial recruitment. Future work will focus on evaluating larger language models, improving logical criteria formalization, and validating the framework using larger collections of real-world clinical documentation.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author(s) used Large Language Models, such as ChatGPT, or similar tools for the refining of the work's contents. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

ACKNOWLEDGEMENTS

The authors would like to thank the Oncology Department of ULS Coimbra for its collaboration and support throughout this work. We gratefully acknowledge Professor Pedro Nuno Furtado for his continuous guidance, insightful feedback, and supervision throughout the development of this research. We also thank Dr. Judy Paulo for her clinical expertise and continuous support during the design and validation of the proposed system, as well as Dr. Margarida Marques and Eng. Adelino Pais for their valuable feedback and contributions, which helped ensure that the proposed solution remained aligned with real clinical needs.

FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

REFERENCES

- [1] N. Bhagat and O. Mackey. Large language models for efficient medical information extraction. *preprint*, n.d.
- [2] B. Gu, V. Shao, Z. Liao, V. Carducci, S. R. Brufau, J. Yang, and R. J. Desai. Scalable information extraction from free text electronic health records using large language models. *BMC Medical Research Methodology*, 25(1):23, Jan. 2025.
- [3] S. Gupta, A. Basu, M. Nievas, J. Thomas, N. Wolfrath, A. Ramamurthi, B. Taylor, A. N. Kothari, R. Schwind, T. M. Miller, S. Nadaf-Rahrov, Y. Wang, and H. Singh. PRISM: Patient Records Interpretation for Semantic clinical trial Matching system using large language models. *npj Digital Medicine*, 7(1):305, Oct. 2024.

- [4] Q. Jin, Z. Wang, C. S. Floudas, F. Chen, C. Gong, D. Bracken-Clarke, E. Xue, Y. Yifan, J. Sun, and Z. Lu. Matching patients to clinical trials with large language models. *Nature Communications*, 15(1):9074, Nov. 2024.
- [5] M. Nieves, A. Basu, Y. Wang, and H. Singh. Distilling large language models for matching patients to clinical trials. *arXiv preprint arXiv:2312.09958*, Dec. 2023.
- [6] V. Ntinopoulos, H. R. Cetina Biefer, I. Tudorache, N. Papadopoulos, D. Odavic, P. Risteski, A. Haeussler, and O. Dzemali. Large language models for data extraction from unstructured and semi-structured electronic health records: a multiple model performance evaluation. *BMJ Health & Care Informatics*, 32(1):e101139, Jan. 2025.
- [7] M. Rahmanian, S. M. Fakhrahmad, and S. Z. Mousavi. Towards efficient patient recruitment for clinical trials: Application of a prompt-based learning model. *preprint*, n.d.
- [8] M. Rybinski, W. Kusa, S. Karimi, and A. Hanbury. Learning to match patients to clinical trials using large language models. *Journal of Biomedical Informatics*, 159:104734, Nov. 2024.
- [9] M. Sushil, V. E. Kennedy, D. Mandair, B. Y. Miao, T. Zack, and A. J. Butte. CORAL: Expert-Curated Oncology Reports to Advance Language Model Inference. *NEJM AI*, 1(4), Mar. 2024.
- [10] D. Vithanage, C. Deng, L. Wang, M. Yin, M. Alkhalaf, Z. Zhang, Y. Zhu, A. C. Soewargo, and P. Yu. Evaluating approaches of training a generative large language model for multi-label classification of unstructured electronic health records. *medRxiv preprint doi:10.1101/2024.06.24.24309441*, Jun. 2024.
- [11] D. Vithanage, P. Yu, Q. Xie, H. Xu, L. Wang, and C. Deng. A comprehensive evaluation of large language models for information extraction from unstructured electronic health records in residential aged care. *Computers in Biology and Medicine*, 197:111013, Oct. 2025.
- [12] M. Wornow, A. Lozano, D. Dash, J. Jindal, K. W. Mahaffey, and N. H. Shah. Zero-shot clinical trial patient matching with LLMs. *arXiv preprint arXiv:2402.05125*, Apr. 2024.